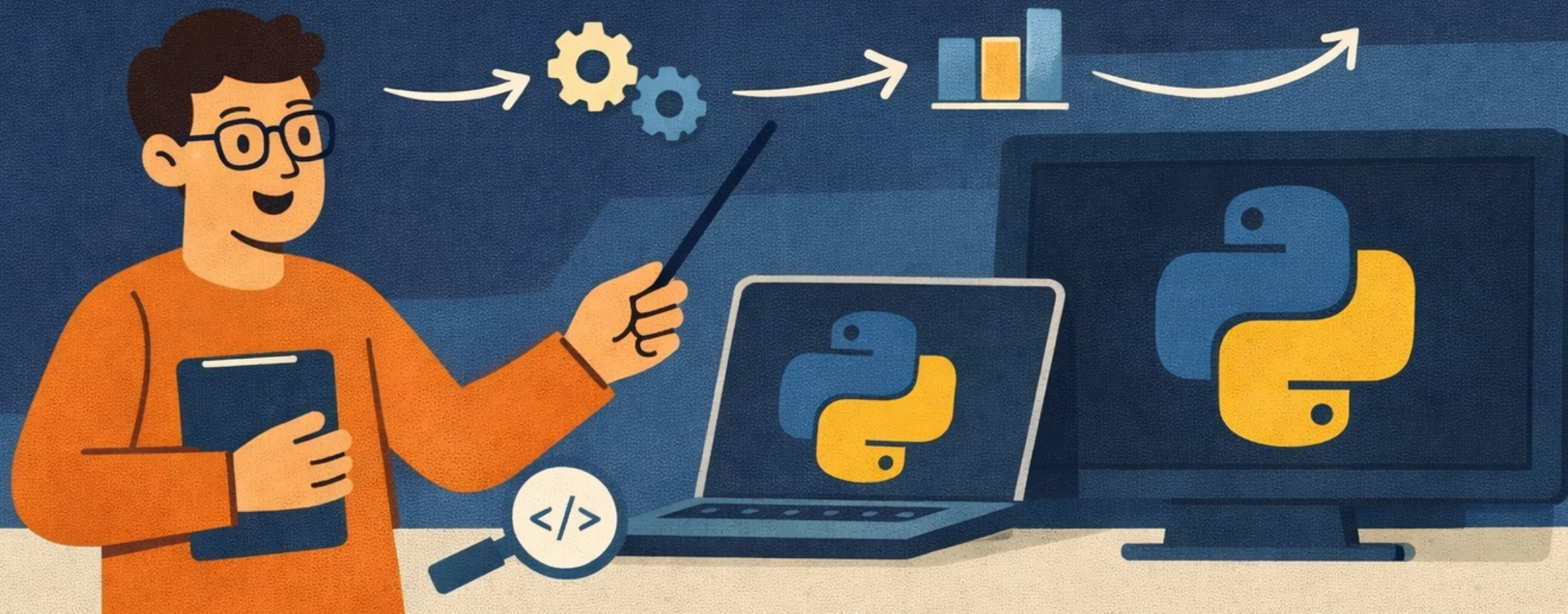


class starts
@ 9:05pm

PYTHON LIBRARIES



Introduction

Aniruddha Mukherjee
(Ani)

Machine Learning Engineer
8 Years of Exp

"Love to
Build & break
things"

"Been with Scaler for 2+ years"

Whats App Group



Link: (Pinned in Notice board)

Purpose: Offline discussion

Building Network

Class updates

peer to peer doubts resolution

Ground Rules

1/ Doubts → Question Tab → "END":
Chat → Interaction / Direct questions

2/ Notes → "Build your own notes"
(My HW notes + Code Files + Scaler material)

3/ Pause → After every sub topic → Clarify doubts

Lecture 1: Introduction to DAV and Numpy

Agenda

- 1) Module Overview
- 2) Introduction to Data Analysis & Visualization (DAV)
- 3) python lists Vs Numpy
- 4) Dimension & Shape
- 5) Indexing.

Module Overview

Numerical
operation

Data
preprocessing

Data
Visualization

✓
"8 lectures"

"python library"

Data Scientist / ML Engineer

python → programming Language

Library (Pandas, Numpy...) → Collection of pre-written code that can be used in your program. Each Library serves a specific task

Module Overview

Bank Manager (Bank Loan App)

	salary	married	gender	prev_loan count	Loan approve
C_1	5L/m	Yes	Male	3	No
C_2	50K/m	Yes	Male	0	Yes

" C_2 " $\rightarrow$ Yes

"Automate this process"

Analysis
machine

→ Finding some hidden pattern within our data

ML context

"Column" → Feature"

Historical
data

$\left\{ \begin{array}{l} 1 \text{ yoe} \rightarrow 10\text{K Salary} \\ 2 \text{ yoe} \rightarrow 20\text{K Salary} \\ 3 \text{ yoe} \rightarrow 30\text{K Salary} \\ \vdots \\ 6 \text{ yoe} \rightarrow 60\text{K Salary} \end{array} \right.$

Pattern

$$\text{salary} = \text{yoe} \times 10\text{K}$$

Exploratory Data Analysis (EDA)

process of Exploring every feature in your dataset before taking a decision.

✓
 S_1

✓
 S_2

✓
 S_3

Exploratory Data Analysis (EDA) ✓✓

pandas

Data Manipulation

Numpy

Num Op

matplotlib / seaborn

Data Visualization

✓✓ Visual → "Interpret pattern"

Why EDA is Necessary?

1/

SL No
mm

1 ✓

2 ✓

3 ✓

4 ✓

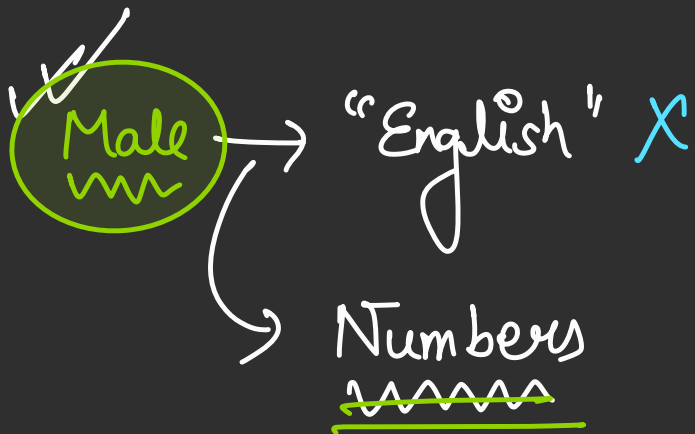
5 ✓

Remove unnecessary columns

2/ Data cleaning



"Making it suitable for ML Model
Training"



Writing Code

→ IDE (Integrated Development Environment)

Pycharm, VSCode, Spyder, Jupyter Notebook . . .

ipynb → python notebook

"Collab Notebook"

→ google

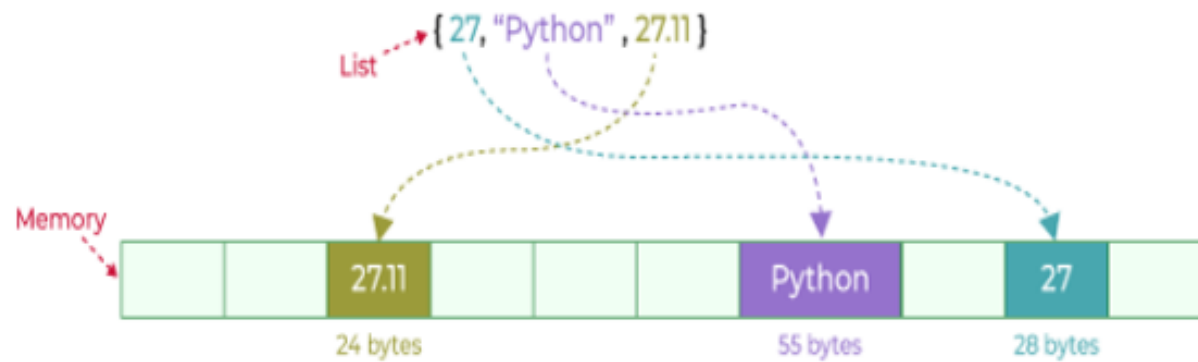
NumPy

↳ Numeric Python

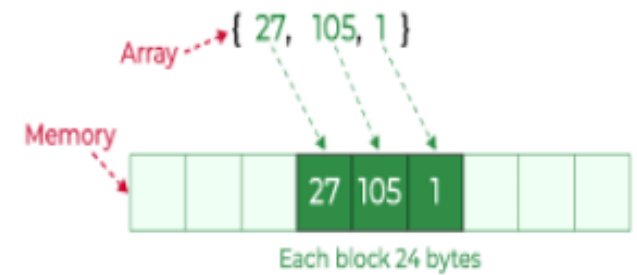
"Numerical operation"

Numpy → Python library
↳ C → "Python wrapper"

Numbers



Python List



NumPy Array

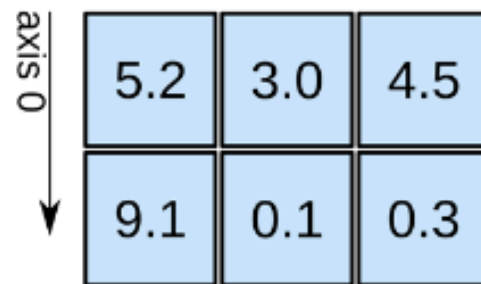
1D array



axis 0 →

shape: (4,)

2D array

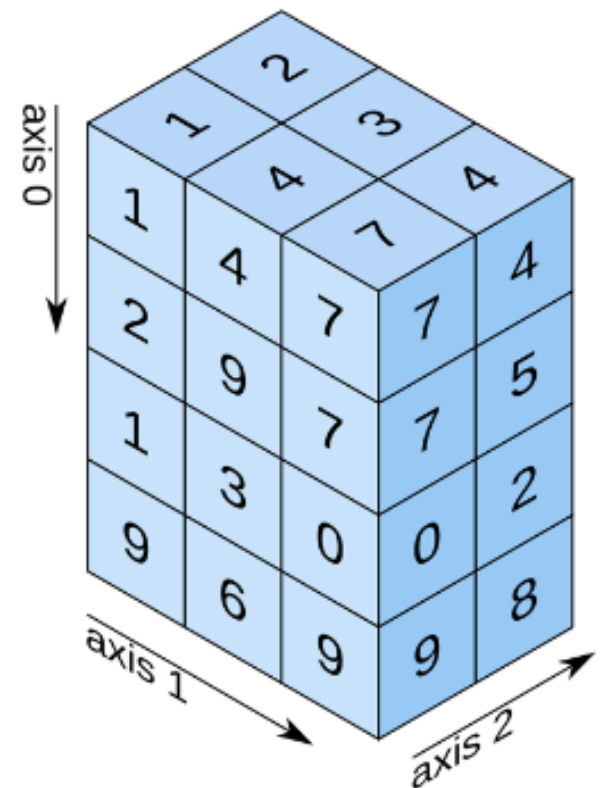


axis 0 ↓

axis 1 →

shape: (2, 3)

3D array



shape: (4, 3, 2)

✓✓ Vscode , Anaconda distribution
~~~~~  
✓✓

(Spyder, Pycharm)  
~~~~~